

On the Energy Costs of Post-Quantum KEMs in TLS-based Low-Power Secure IoT

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ABSTRACT

Recent achievements in designing quantum computers place a serious threat on the security of state-of-the-art public key cryptography and on all communication that relies on it. Meanwhile, security is seen as one of the most critical issues of low-power IoT devices even with pre-quantum public key cryptography since IoT devices have strict energy constraints and limited computational power. Thus, state-of-the-art dedicated hardware accelerators have been deployed to facilitate secure and confidential communication with well established protocols on such devices.

It is common belief that the complexity of the cryptographic computations are also the bottleneck of the new, quantum-resistant algorithms and that hardware accelerators are necessary to use them efficiently on energy constrained embedded devices. In this paper, we carried out an in-depth investigation of the application of potential Post-Quantum Cryptography algorithms, which were proposed in the associated US NIST process, to a representative TLS-based low-power IoT infrastructure.

First, we show that the main contributor to the TLS handshake latency are the higher bandwidth requirements of post-quantum Key-Encapsulation Mechanisms rather than the cryptographic computations itself. Second, from the perspective of crypto-agility we show that edge devices with code-based, isogeny-based as well as lattice-based algorithms have low energy consumption, which enables long battery run times in typical IoT scenarios without dedicated hardware accelerators. Third, we increase the level of security further by combining pre-quantum and post-quantum algorithms to a hybrid key exchange, and quantify the overhead in energy consumption and latency of it.

CCS CONCEPTS

• **Hardware** → **Power estimation and optimization**; • **Computer systems organization** → *Sensors and actuators*; • **Security and privacy** → **Public key encryption**.

KEYWORDS

Post-Quantum Cryptography, low-power secure IoT, Key-Encapsulation Mechanisms

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1 INTRODUCTION

Industry 4.0 and its trend to exchange, collect and process data to enhance industrial processes (the *Industrial Internet of Things* (IIoT)) have led to the fact that safety-critical, infrastructural applications as well as industrial secrets are reliant on the security of the communication [25]. The established standard for secure communication over the internet is *Transport Layer Security* (TLS), whose security relies on the combination of public key and symmetric cryptography.

The state-of-the-art public key algorithms are hard to break on classical computers, but their underlying mathematical problems can be solved in polynomial time with *Quantum Computers* (QCs), resulting in a complete break of the cryptosystems as soon as powerful QCs are available [37]. Thus, the US *National Institute of Standards and Technology* (NIST) started a *Post-Quantum Cryptography* (PQC) standardization process to find new, quantum computer-resistant public key primitives for *Digital Signature Algorithms* (DSAs) and *Key-Encapsulation Mechanisms* (KEMs) [32]. Whether or not large scale QCs will be available in future, it is generally agreed that PQC will be the new standard cryptography [15]. While both DSAs and key exchange cryptosystems can be broken with powerful QCs, the urgency of migrating to post-quantum KEMs is arguably higher, as attackers can harvest data today by eavesdropping and decrypt it as soon as they have access to a QC ("store now, decrypt later" [15]).

Due to limited computational capability and strict energy constraints of battery-powered IoT edge devices on the one hand and

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the computational intensive state-of-the-art public key algorithms on the other hand, dedicated *Cryptographic Hardware Accelerators* (CHAs) are today deployed to facilitate secure communication even with computationally complex protocols like TLS. Since the CHAs have high energy efficiency, they enable years of battery life-time in typical secure low-power IoT applications [25]. It is common belief that dedicated hardware accelerators are also a necessity for the efficient implementation of PQC on energy constrained embedded devices [16] [6] [30].

To reevaluate this necessity, we explore in this paper the overhead in energy demand and latency caused by the migration from conventional to post-quantum key exchange in a representative low-power IoT infrastructure which employs TLS, *IPv6 over low-power Wireless Personal Area Network* (6lowPAN), *Transmission Control Protocol* (TCP) and *Message Queuing Telemetry Transport* (MQTT). We extended mbedTLS¹, a standard TLS library for embedded systems, to allow for the seamless integration of post-quantum KEMs and hybrid key exchanges into the TLS 1.2 handshake protocol. As the security of individual KEMs is not certain yet due to their novelty, we examine algorithms of all three problem families that are part of the NIST process to determine their individual costs in the view of crypto-agile systems. The implementation assures *Perfect Forward Secrecy* (PFS) for all KEMs by using ephemeral keys.

Through this exploration, we respond to NIST's recent call for more performance data of PQC applications to internet protocols [32]. We also include the more extensive FrodoKEM into our investigations, which has been recommended by the German Federal Office for Information Security for high security applications [15]. The key contributions of this paper are:

- We provide a holistic analysis of the latency and energy demand of the TLS Handshake for selected NIST Round 3 KEMs and conclude that the bandwidth requirements (the sum of public key and ciphertext) rather than the cryptographic computations are the main bottleneck of code-based and lattice-based KEMs.
- We show that KEMs of all problem families can efficiently be implemented into a representative low-power IoT setup, yielding low energy demand as well as long life times for battery-powered edge devices.
- We explore the overhead in latency and energy demand caused by hybrid-key exchanges to fulfill strict governmental security requirements [15].

To the best of our knowledge, this is the first work which provides a broad system view of the application of PQC to a representative, low-power IoT application. This paper is structured as follows. In the next section, we expound the background of TLS, explain how TLS can be broken with QCs and highlight the fundamentals of PQC. In section three, an overview of related work regarding low-power secure IoT, post-quantum TLS implementations, benchmarking and the development of dedicated CHAs for PQC primitives is provided. In section four, our IoT evaluation platform is presented. In section five, the measurement results are shown and evaluated. In section six, we summarize our key-observations.

2 BACKGROUND

This section covers the background of TLS, how its security can be broken with QCs and how this problem can be solved with PQC.

2.1 TLS

TLS is a standardized cryptographic protocol for client/server applications to secure communication against eavesdropping, tampering and message forgery [14]. This is ensured by utilizing a combination of multiple symmetric and public key cryptosystems. Data privacy is provided through symmetric encryption of the application data and data integrity through *Message Authentication Codes* (MACs). Initially, a master secret from which the keys for the symmetric primitives are derived is established and authenticated with public key cryptography.

TLS consists two different layers, the TLS Record Protocol and the TLS Handshake Protocol. The record protocol is placed underneath the handshake protocol and cares for data (de-)fragmentation, (de-)compression, transfer and is responsible for encryption and decryption of the application data.

The TLS Handshake Protocol is used to initially negotiate the session, including the authentication of the communication partners, establishing a common master secret and the instantiation of security parameters. Its principal is shown in figure 1 and it is explained in the following with focus on the authenticated key exchange.

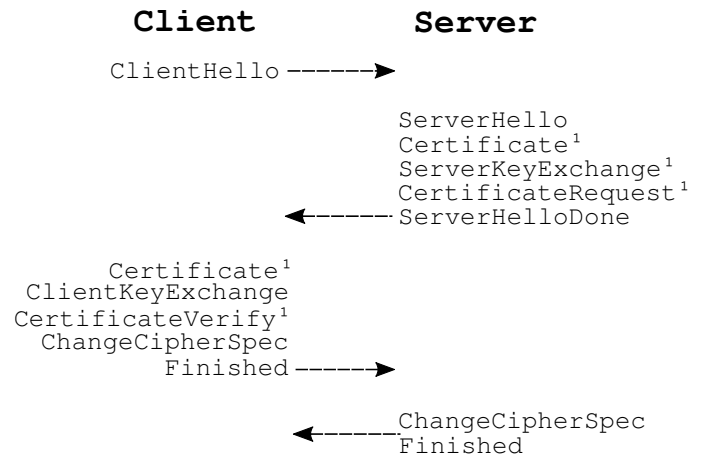


Figure 1: TLS 1.2 Handshake protocol as defined in [14]. The messages marked with ¹ are optional handshake messages.

First, the client and server negotiate a common ciphersuite in the ClientHello and ServerHello messages which determines the cryptographic algorithms for key exchange, digital signature, application data encryption and the hashing function. Then, the ServerCertificate message conveys the server's certificate chain, including its public key p_k , to the client. For some key exchange mechanisms (e.g. *Elliptic Curve Diffie-Hellmann key exchange in ephemeral mode* (ECDHE)), the client requires additional information and the certificate is not sufficient to calculate the premaster secret pms . Here, the server creates an ephemeral set of keys (d_s ,

¹<https://github.com/ARMmbed/mbedtls>

P_S), signs the public key P_S using the DSA specified in the cipher-suite to prevent man-in-the-middle attacks, and sends them in the optional ServerKeyExchange message. Next, the client answers in the ClientKeyExchange with its corresponding data that is necessary for the server to calculate the same pms. In case of ECDHE, the client also generates an ephemeral set of keys (d_C , P_C) and transmits the public P_C to the server. The server calculates pms based on d_S , P_C , P_S while the client calculates pms based on d_C , P_C , P_S .

Finally, the ChangeCipherSpec message is sent by both server and client to signalize that further messages will be encrypted based on a shared secret derived from pms. The Finished message is the first encrypted message based on pms to verify the correctness of the authenticated key establishment.

2.2 Breaking TLS with Quantum Computers

The security of state-of-the-art public key algorithms is based on the intractability of the *Discrete Logarithm Problem* (DLP) (e.g. in case of elliptic curve cryptography) or the hardness of large integer factorization (in case of *Rivest-Shamir-Adleman* (RSA) cryptosystem). These hardness assumption are threatened by the recent advancements in building QCs which work fundamentally different than conventional computers. Instead of operating on bits with the two distinct states 0 and 1, QCs base on *qubits* which can be in a superposition of these two. With quantum gates rather than standard logic gates this superposition and further quantum mechanic effects like quantum entanglement and wave function collapse can be used to efficiently solve certain mathematical problems which are practically intractable on conventional computers.

The quantum algorithm to solve the DLP and large integer factorization in polynomial time has already been developed by Peter Shor in 1994 at the Bell Labs [37], however, cryptographically relevant QCs have been out of reach back then. Meanwhile, this has changed as estimations state that by 2031 large scale QCs could break today's cryptosystems with a 50% chance [29].

The vulnerability of the public key algorithms in TLS can be used in two ways by QC-based attacks. First, Shor's algorithm can be executed in a passive attack, e.g. by eavesdropping on one of the public keys in ServerKeyExchange or ClientKeyExchange of ECDHE. Then, the attacker retrieves the secret key (either d_C or d_S in our example) by applying Shor's algorithm on one of the public keys and calculates the same pms as client and server, resulting in a break of privacy. From then on, all application data sent between server and client can be decrypted by the eavesdropper.

Second, Shor's algorithm can be used in a *man-in-the-middle* (MITM). The MITM can impersonate as server to the client and as client to the server and secretly forward their messages. From the server's certificate, the attacker can retrieve the server's secret key with Shor's Algorithm. From then on, the attacker can forge the server's signature and use it to alter messages (e.g. by providing its own public key to the client, but signed with the server's signature), thus resulting in an entire break of the TLS implied security until the server's certificate is expired. In critical infrastructures, this can maliciously be used to send false commands and, in the worst-case scenario, lead to system failures.

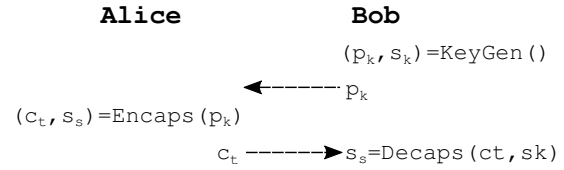


Figure 2: Simple KEM-based key exchange Protocol

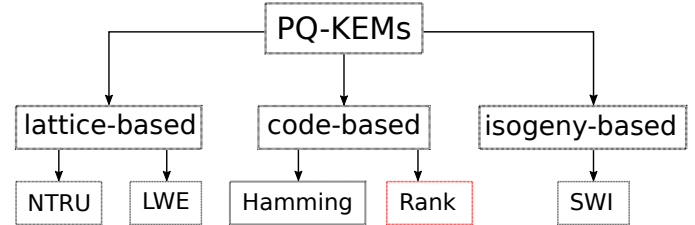


Figure 3: Overview of the KEM-problems used in the NIST process

2.3 PQC

Due to the vulnerabilities of state-of-the-art cryptography PQC has emerged to find new, QC resistant public key cryptosystems. Even though both DSAs and key exchange mechanisms can be broken, the urgency of migrating from classical cryptography to PQC is different in IoT applications. While forging signatures offers the attacker more extensive possibilities, it initially requires access to a QC. Key exchange mechanisms on the other hand can be broken by a passive attacker even years after the communication occurred ("store now, decrypt later" [15]). Attackers can start eavesdropping on sensible encrypted data today and decrypt it as soon as they have access to a powerful QC. Further typical factors that contribute to this urgency are product life-cycles and the difficulty of in-field software updates for IoT devices. Therefore, we focus on the application of post-quantum KEMs in this paper.

KEMs are a set of the three algorithms (KeyGen, Encaps, Decaps) that form a public key cryptosystem. Unlike general purpose public key encryption schemes, which can be used to encrypt application data, KEMs serve the sole purpose of establishing a shared secret between communication partners (figure 2). Similar to standard ECDHE key exchange, Bob initially creates a set of two keys, a secret key s_k and a public key p_k in KeyGen. Alice takes the public key p_k and calculates the ciphertext c_t and shared secret s_s with the Encaps function. With the c_t , s_k and the Decaps function, Bob calculates the same s_s .

The US NIST started a standardization process in 2016 to find new PQC methods, which is divided into multiple rounds [31]. In the recent, third round four finalists and five alternate KEMs have been elected as possible future standards [2]. Those algorithms do not rely on the hardness of integer factorization or the discrete logarithm, but rely either on problems over lattices, codes, or on isogenies, see figure 3. Up to today, those problems are assumed to be secure against QCs.

2.3.1 Lattice-based KEMs. The lattice-based KEMs rely either on derivations of the Learning-with-Errors (LWE) problem or use the

constructions of the NTRU cryptosystem. NTRU has been introduced in 1998 by Hoffstein et al. [20] and built the basis for various cryptosystems. Therefore, it has been analyzed for a long time and provides a strong confidence of the underlying security assumptions. The idea of the original NTRU-scheme is used in two KEMs in the third round, the finalist NTRU [11] and the alternate candidate NTRUPrime [9].

The LWE problem was first introduced by Regev in 2005 [34] and has since then received great attention in cryptography. In the NIST process, two finalists, SABER [13] and KYBER [5], are based on derivations of the LWE problem, while FrodoKEM relies on the hardness of the plain LWE problem [3]. The *structured lattice-based KEMs* (SLB-KEMs) NTRU, SABER and KYBER are considered the most promising candidates for standardization because they offer good performance with small ciphertexts and public keys. Still, the alternate candidate FrodoKEM is less vulnerable to algebraic attacks as its lattice is less structured and is recommended by the German Federal Office for Information Security [15].

2.3.2 Code-based KEMs. Code-based cryptography was introduced by McEliece in 1978 and is built from error-correction codes [26]. McEliece's original approach uses a hidden error-correcting binary Goppa code whose generator matrix is multiplied with two randomly selected, secret matrices to build the public key. The fourth finalist, Classic McEliece, is based on this construction [8] while the alternate candidates BIKE [4] and HQC [1] rely on hard problems over quasi-cyclic codes to reduce the bandwidth requirements.

The before mentioned code-based KEMs all rely on the hamming-metric. The NIST submissions also featured two KEMs which use the rank-metric instead, ROLLO [27] and RQC [28]. Due to efficient algebraic attacks on rank-metric codes in [7] those KEMs were not further considered for standardization despite adjusting their parameter-sets to the newly discovered attacks.

Code-based KEMs have larger keys compared to SLB-KEMs, with the rank-metric ROLLO_I being an exception from this. Nonetheless, they provide completely different security assumptions than lattice-based KEMs and are valid alternatives in case that the SLB-KEMs became insecure.

2.3.3 Isogeny-based KEM. Only one KEM, the last alternate candidate SIKE [21], is based on the supersingular isogeny walk problem in the NIST process. While SIKE has clearly been shown to have very expensive calculations, e.g. in [23], it is equipped with the lowest bandwidth requirements and provides a unique security assumption to the pool of KEMs.

Table 1: NIST security levels in the PQC standardization process as defined in [31].

Security Level	Minimum Security Requirements
1	AES-128
2	SHA-256
3	AES-192
4	SHA-384
5	AES-256

NIST uses five different security levels in their initial call for post-quantum primitives, see table 1. The minimum security requirements define how much computational resources are required to successfully break the cryptosystem, for example, attackers need at least the same resources as for a key search on *Advanced Encryption Standard-128* (AES-128) to break NIST level 1 KEMs [31].

2.4 Hybrid key exchange

Even though the PQC primitives receive great worldwide attention and cryptanalysis, some of their security assumptions are quite new and require further research, including vulnerabilities to classical instead of quantum computers. Therefore, hybrid key exchange is recommended as an intermediate step before fully migrating to PQC [15].

Hybrid key exchange combines the security of PQC and conventional key exchange like ECDHE, to provide at least the same security as in the pre-quantum era. The central idea is to use both key exchange mechanisms to agree on two different premaster secrets instead of one, combine them (e.g. by concatenation or the XOR-operation [12]) and then use a key derivation function to process the master secret.

3 RELATED WORK

This section provides an overview of recent research activities in TLS-based low-power IoT security, the integration of PQC primitives into TLS and the acceleration and benchmarking of the NIST post-quantum-KEMs.

3.1 IoT Security

Security has been valued as one of the key challenge for the future of IoT. The usage of TLS to protect the IoT communication against eavesdropping, tampering and message forging has been pointed out in [17] [19] [18], however, its application has been viewed critically due to the computational requirements. A solution for this problem has been provided in [25] and now, feasible implementations yield years of battery life time even for complex protocols like TLS. Optimizations on the network system have shown to significantly improve the energy efficiency of TLS-based low-power IoT infrastructures in [24].

3.2 Post-Quantum TLS

Open Quantum Safe (OQS) is one of the most advanced open-source projects for simplifying the prototyping of PQC [38]. In this project, post-quantum OpenSSL implementations¹ have been developed for TLS 1.2 and TLS 1.3. A network emulation framework was developed in [33] which uses the OQS implementations to assess on the trade-offs of PQC in TLS. They showed that in fast reliable networks the TLS handshake completion time is mostly determined by the public key cryptography, while in networks with higher packet loss rates the communication has a stronger influence on completion time. Crockett et al. conducted case studies to explore the challenges and design space of implementing post-quantum and hybrid key exchange into TLS 1.2 and TLS 1.3 and observed that the limited message size in TLS 1.2 might be exceeded by

¹<https://github.com/open-quantum-safe/openssl>

some KEMs [12]. The only post-quantum TLS implementation on embedded devices until now was recently presented in [10]. The authors integrated one parameter set of the post-quantum KEM KYBER and the DSA SPHINCS+ into the mbedTLS library and conducted measurements on three different devices, where the devices acted either as server or client. They concluded that TLS handshakes with those two schemes are feasible on their devices, however, they omitted the impact of the communication overhead in their measurements, observed only one KEM and did not provide information on the energy demand.

3.3 Acceleration of PQC primitives

The pqm4 project developed a library with assembler optimized versions of some post-quantum KEMs for Cortex M4 based controllers [22]. Based on this project, Kannwischer et al. performed benchmark tests and showed that they can achieve significant speed ups for most KEMs through their optimizations. The authors in [35] measured the energy demand of the post-quantum KEM calculations based on the same library and showed that SLB-KEMs have the lowest energy demand out of all NIST proposals.

Dedicated CHAs have been developed for most KEMs proposed in the NIST process, however, the SLB-KEMs are among those which received the greatest attention. In [6], a co-processor was developed and fabricated as hardware accelerator for FrodoKEM, KYBER and NewHope, whereas NewHope is a SLB-KEM which was removed in favor of KYBER in the third NIST round. They increased the energy-efficiency of those KEMs by an order of magnitude, thus making the implementation of post-quantum KEMs on low-power embedded devices more feasible. In [16], a custom instruction-set extension was designed to improve the performance of the lattice-based schemes KYBER, NewHope and SABER on low cost resource constrained RISC-V processors. Their ASIC implementation reduced the energy demand of the schemes by a factor of 9.5. Nejatollahi et al. designed a hardware accelerator to improve the energy efficiency of the *Number Theoretic Transform* (NTT) used inside of NewHope and KYBER for IoT applications [30]. They reduced the energy demand by a factor of 1.9.

To the best of our knowledge, no work has provided an in-depth investigation of the costs in terms of energy and latency of post-quantum KEMs in TLS based low-power secure IoT yet. Therefore, the extent to which the application of PQC to typical low-power IoT setups is feasible without dedicated CHAs is still unknown. We present a holistic analysis of such a representative application in this paper, supporting NISTs call for more performance data of PQC in protocol based applications [32].

4 IOT SETUP

This section describes the IoT infrastructure which is used for the evaluation of post-quantum KEMs. Furthermore, the reasoning for exploring the respective KEMs is explained.

4.1 Low-Power IoT Evaluation System

The low-power IoT infrastructure in this paper is based on [24], see figure 4, and is designed to provide TLS-based end-to-end encryption. The gateway is a Raspberry Pi 3 and acts as a headless router, i.e. it forwards the communication from the *Bluetooth Low Energy*

(BLE) physical layer to the Ethernet physical layer and vice versa without affecting the *Internet Protocol* (IP)-Layer. For a trade-off between connection handling, data throughput and energy demand a BLE connection interval value of 20ms has been chosen [24].

The server is a standard Linux server with a MQTT broker and an Intel(R) Xeon(R) Silver 4214 @ 2.2GHz which features the Advanced Vector Extension (AVX) instructions. The battery-powered IoT edge devices employ the nRF52840 *System on Chip* (SoC) by Nordic Semiconductor as main computational unit, which incorporates an ARM Cortex M4 *Micro Controller Unit* (MCU), a security subsystem (ARM CryptoCell-310) and a Bluetooth 5-compliant 2.4 GHz radio. The MCU has 1MB of flash storage and 256KB *Random-Access-Memory* (RAM). The security subsystem provides low-power hardware acceleration for *Elliptic-Curve DSA* (ECDSA) and ECDHE key exchange. It also comes with hardware accelerators for AES128 and SHA2.

The NIST curve p-256 is used for all elliptic curve cryptography based operations in this setup. The public key is signed using the conventional ECDSA algorithm. Ephemeral keys are generated on every new handshake to ensure PFS.

The current consumption of the edge device was recorded with a Keithley DMM7510 digital multimeter. For our measurement, we used a sampling rate of 20kHz and 0.1μA resolution. The latency was measured and synchronized with the current measurements by using a high speed logic analyzer.

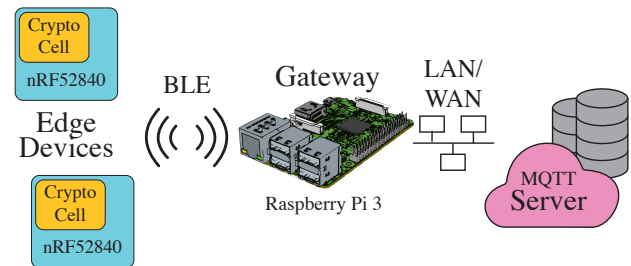


Figure 4: Network Topology of the evaluated IoT System [24]

4.2 Software Implementation

TLS-based security with post-quantum key exchanges was implemented on server and edge devices. Regarding the gateway, no adjustments were required as it only interferes in the communication on the physical layer.

4.2.1 Consideration and Implementation of KEMs. At a first instance, all KEMs were considered for implementation which are currently part of the NIST process ("finalists" and "alternate candidates"). Against the background of the application-specific resource constraints the following metric was used to evaluate the KEMs:

- **Bandwidth Requirements and RAM usage:** In the TLS handshake the public key p_k and the ciphertext c_t are exchanged and both have to be stored in the RAM at the same

time. Classic McEliece exceeded the available memory resources of the state-of-the-art MCU on the SoC. RAM requirements during computation are a further limiting factor and have been exceeded by HQC and FrodoKEM for higher security levels. Therefore, those KEMs have only been considered in security level 1.

- **Software Implementations:** Reference software implementations are part of the NIST submissions of every KEM. Even though some KEMs have already been optimized for the Cortex M4 [22], the reference implementations were used to allow an unbiased comparison to those KEMs which are not optimized yet. FrodoKEM640 and SIKE are exceptions from this, as their reference implementations are not feasible for low-power IoT setups. BIKE and NTRUPrime include external libraries, e.g. OpenSSL or the *Number Theory Library* (NTL), which cannot be directly ported to the target MCU. Therefore, they have not been considered in this work.
- **Wide-spread hardness assumptions:** The rank-metric code-based cryptosystems RQC and ROLLO were also included to explore distinct problem families despite the removal from the standardization process. Rank-metric cryptography has some unique properties, e.g. ROLLO-I is the only code-based cryptosystem which has comparable bandwidth requirements like SLB-KEMs. Due to the implementation overhead, only the level-1 parameter sets of those KEMs were considered in this investigation.

Table 2 shows the KEMs which we considered. All parameter-sets of the SLB-KEMs were implemented to determine the overhead in energy and latency for higher security levels, as one of them is very likely to be standardized [32]. Despite its removal from the process, the SLB-KEM NewHope has been implemented, as many works designed dedicated hardware accelerators for it.

On top of their algorithm-specific computations, KEMs also use hashing-functions (SHA2 and SHAKE) and block-ciphers (AES) as building blocks, e.g. as *Extendable Output Function* (XOF), inside their calculations. Cortex M4 assembler optimized implementations have been used for SHAKE from the Keccak Team¹ and the Cortex-M4 optimized implementations by pqm4 from [22] for AES192 and AES256. For AES128 and SHA2 the hardware accelerator of the target SoC security subsystem was used.

4.2.2 Edge devices. The OpenSSL implementation of TLS requires too much memory for typical embedded devices. Therefore, the post-quantum TLS version of the OQS project cannot be used. Instead, a custom post-quantum extension of the open-source mbedTLS library was implemented.

For compatibility, this work follows the same approach as the OQS project. Therefore, the *ServerKeyExchange* message and the *ClientKeyExchange* messages are used to transmit the public key and ciphertext of the KEM.

4.2.3 Server. The open-source Eclipse Mosquitto is used as MQTT broker² on the server, which provides an *Application Programming Interface* (API) for the integration of OpenSSL. The TLS 1.2 version

of the OQS project, OQS-OpenSSL_1_0_2³, was used to implement post-quantum security on the server.

5 RESULTS

This section covers the measurement results and their interpretation of the post-quantum KEM-based TLS handshakes. First, the major causes for the impact of post-quantum KEMs on the TLS handshake latency and energy demand are determined. Then, post-quantum KEM-based handshakes are evaluated and compared with classical and hybrid key exchanges in terms of energy and latency. Based on these results, we determine the battery life times of edge devices for the different KEMs.

The following measurements have been carried out ten times for each KEM. The mean values of the results have been used for the figures and the standard deviation is displayed with error bars.

5.1 Latency and Energy

The TLS handshake latency and energy demand have been measured to identify the bottleneck of the KEMs and to evaluate their trade-offs. The handshaking procedure was recorded between sending the *ClientHello* message and receiving the *Finished* message from the Server.

5.1.1 Latency. Figure 5 shows the TLS 1.2 handshake latencies with NIST security level 1 KEMs and with the hardware-accelerated pre-quantum ECDHE. The orange colored-bar represents the time period in which the edge device performs the KEM's *Encaps*-function. The blue bar indicates the time during which the radio subsystem is transmitting c_t to the server.

The grey bar represents the handshake time minus the orange bar and the blue bar. This includes receiving and sending the certificates (roughly 2KB of data), receiving the KEM's p_k , waiting for the server responses, signing, verifying, data processing and the remaining TLS handshake steps. We observed that the server's calculations of the KEM (KeyGen and Decaps) have only a minor influence on the grey bar.

The chart shows that, except for SIKE, the cryptographic computations in *Encaps* have a negligible influence on the latency. This is contrary to the observations in pre-quantum handshakes. There, the cryptographic calculations have been shown to be the major contributor to the latency with software-based ECDHE [25]. Another significant contributor to the overall latency in handshakes with post-quantum KEMs is the time demand to exchange c_t and p_k , i.e. the KEM's bandwidth requirement. This is substantiated by the trend line of the bandwidth requirements in figure 6, as the relation between the bandwidth requirements of the respective KEMs shows the same trend as the communication in figure 5 (blue and grey bars).

This leads to the conclusion that the latency overhead of post-quantum KEMs is predominantly determined by the bandwidth requirements rather than the computation.

5.1.2 Energy. The energy demand of TLS handshakes is shown in figure 7. Handshakes with the SLB-KEMs and the rank-metric ROLLO require the least energy, which is also slightly smaller than that of ECDHE because of their faster computations. SIKE has the

¹<https://github.com/crystalsnetworkdev/pq4>

²<https://github.com/eclipse/mosquitto>

³<https://github.com/open-quantum-safe/openssl>

Table 2: Consideration of the NIST KEMs for implementation in this paper. The NIST Level shows the NIST security level according to table 1; $|p_k|$ and $|c_t|$ denote the size of the public key and ciphertext in bytes. The competition status indicates if the KEMs are still considered in the third round of the NIST standardization process.

Name	Competition Status	Problem Family	NIST Level	$ p_k $	$ c_t $	Implementation
mceliece348864	Finalist	code	1	255K	128	no (bandwidth)
mceliece460896	Finalist	code	3	512K	188	no (bandwidth)
mceliece6688128	Finalist	code	3	1M	240	no (bandwidth)
mceliece6960119	Finalist	code	5	1M	226	no (bandwidth)
mceliece8192128	Finalist	code	5	1.3M	240	no (bandwidth)
KYBER512	Finalist	lattice	1	800	736	Reference-C
KYBER768	Finalist	lattice	3	1184	1088	Reference-C
KYBER1024	Finalist	lattice	5	1568	1568	Reference-C
NTRUHP52048509	Finalist	lattice	1	699	699	Reference-C
NTRUHP52048677	Finalist	lattice	3	930	930	Reference-C
NTRUHRSS701	Finalist	lattice	3	1138	1138	Reference-C
NTRUHP54096821	Finalist	lattice	5	1230	1230	Reference-C
LIGHTSABER	Finalist	lattice	1	672	736	Reference-C
SABER	Finalist	lattice	3	672	1088	Reference-C
FIRESABER	Finalist	lattice	5	1312	1472	Reference-C
FrodoKEM640	Alternate	lattice	1	9616	9720	PQM4 [23]
FrodoKEM976	Alternate	lattice	3	15632	15744	no (RAM usage)
FrodoKEM1344	Alternate	lattice	5	21520	21632	no (RAM usage)
HQC128	Alternate	code	1	3040	6017	Reference-C
HQC192	Alternate	code	3	5690	11364	no (RAM usage)
HQC256	Alternate	code	5	8698	17379	no (RAM usage)
BIKE_L1	Alternate	code	1	1541	1573	no (external libraries)
BIKE_L3	Alternate	code	3	3083	3115	no (external libraries)
ntrupr653	Alternate	lattice	2	897	1025	no (external libraries)
ntrupr761	Alternate	lattice	3	1039	1167	no (external libraries)
ntrupr857	Alternate	lattice	4	1184	1312	no (external libraries)
SIKE_P434	Alternate	isogeny	1	346	330	based on [36]
SIKE_P503	Alternate	isogeny	2	378	402	based on [36]
SIKE_P610	Alternate	isogeny	3	462	486	based on [36]
SIKE_P751	Alternate	isogeny	5	564	596	based on [36]
RQC-I	removed	code	1	1834	3652	Reference-C
RQC-II	removed	code	3	2853	5690	no (implementation overhead)
RQC-III	removed	code	5	4090	8164	no (implementation overhead)
ROLLO-I-128	removed	code	1	696	696	Reference-C
ROLLO-I-192	removed	code	3	958	958	no (implementation overhead)
ROLLO-I-256	removed	code	5	1371	1371	no (implementation overhead)
ROLLO-II-128	removed	code	1	1941	2089	Reference-C
ROLLO-II-192	removed	code	3	2341	2469	no (implementation overhead)
ROLLO-II-256	removed	code	5	2559	2687	no (implementation overhead)
NewHope512CCA	removed	lattice	1	928	1120	Reference-C
Newhope1024CCA	removed	lattice	5	1824	2208	Reference-C

highest energy demand despite its moderate handshake latency. This indicates that extensive calculations have a larger impact on the energy demand than high bandwidth requirements. Likewise,

FrodoKEM640-AES requires less energy although its transmission time is longer than SIKE's calculation time.

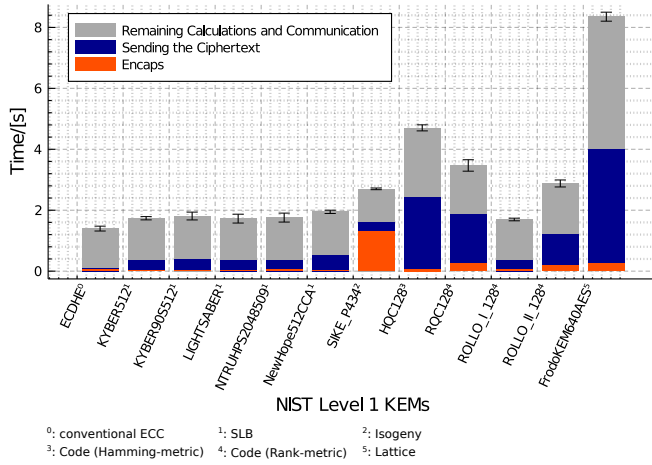


Figure 5: TLS 1.2 handshake latency for NIST level 1 categorized KEMs with 20ms BLE Connection Interval

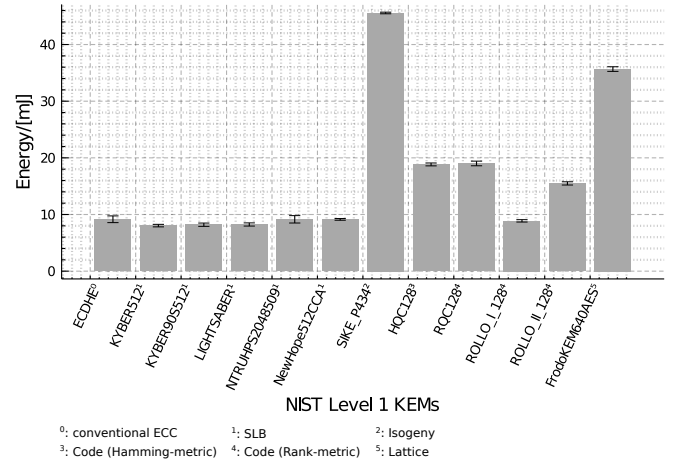


Figure 7: TLS 1.2 handshake energy demand for NIST level 1 categorized KEMs with 20ms BLE Connection Interval

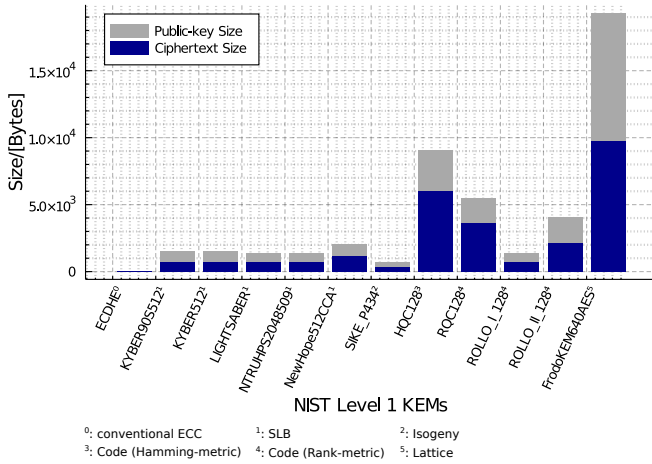


Figure 6: Bandwidth requirements of the NIST level 1 categorized KEMs

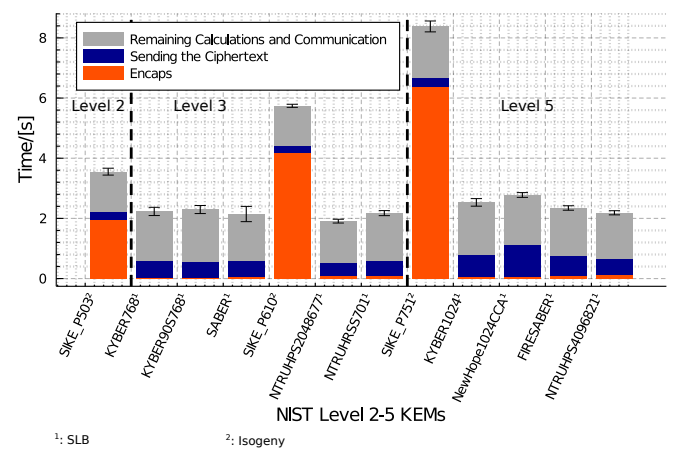


Figure 8: TLS 1.2 handshake latency for NIST level 2-5 categorized KEMs with 20ms BLE Connection Interval

5.1.3 Overhead of higher security levels. Figure 8 and 9 show the results for higher security KEMs, where the vertical lines mark transitions from security level 2 to 3 and from 3 to 5. The observations made in the NIST level 1 results also apply here: Handshakes with SLB-KEMs in the highest NIST level have a lower latency and a smaller energy demand than that with code-based and isogeny-based KEMs on NIST level 1. KYBER's latency and energy demand increases by 25% in level 3 and by 45% in level 5 compared to level 1.

5.1.4 Influence of BLE connection interval. The connection interval of BLE is an important parameter, which defines how frequent both peers are exchanging data. Thus, it is the determining factor for the overall throughput of BLE. In order to test how these observations are affected by this parameter, the same measurements have been carried out for the fastest possible BLE connection interval of 7.5ms.

Figure 10 shows these results. The chart shows that the handshake latency is reduced by the same 60% as the BLE connection interval reduction from 20ms to 7.5ms, which indicates a linear dependency. Besides that, the difference in latency between ECDHE and LSB-KEMs is reduced to less than 5%. Hence, the previous observations can also be applied for smaller BLE intervals and Encaps has still little influence on the latency for most KEMs.

In general, smaller BLE connection intervals increase the energy demand of a device because the radio is active more frequently once a connection is established, regardless of the need for actual data transmissions. However, in our application it also decreases the time in which the MCU is in an active state due to the higher throughput. For HQC, RQC and FrodoKEM, it was observed that these two effects compensate each other and only negligibly affect the energy demand of handshakes when decreasing the connection

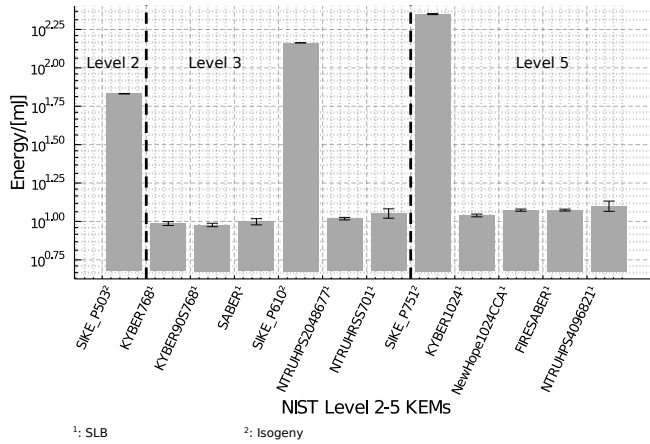


Figure 9: TLS 1.2 handshake energy demand for NIST level 2-5 categorized KEMs with 20ms BLE Connection Interval

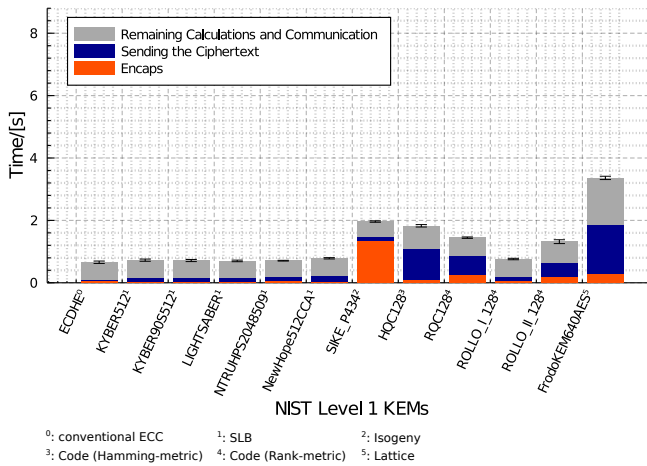


Figure 10: TLS 1.2 handshake latency for NIST level 1 KEMs with 7.5ms BLE Connection Interval

interval from 20ms to 7.5ms. For KEMs with lower bandwidth requirements and fast computations (SLB-KEMs and ROLLO_L) and ECDHE the energy demand is reduced by 20%. The energy demand of handshakes with SIKE, whose time demand is dominated by computation, experiences a contradictory behaviour. Here, the increased energy demand by the radio surpasses the potential energy savings induced by the higher throughput, leading to an increase in energy demand of 2%.

5.2 Hybrid key exchange

In the hybrid-key exchange, ECDHE is executed on top of the KEM, which increases the energy demand, memory requirements and latency. To minimize these costs, the CHA in the SoC security subsystem has been used for the elliptic-curve related calculations.

Figure 11 shows relative overhead for hybrid-key exchange compared to non-hybrid mode for the NIST level 1 finalists and alternate candidates.

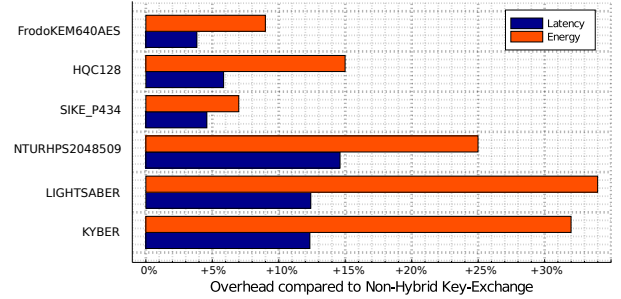


Figure 11: Overhead in latency and energy for hybrid key exchanges in the TLS handshake compared to non-hybrid post-quantum KEMs with 20ms BLE Connection Interval

SLB-KEMs have the largest relative overhead; the latency increases on average about 13% and the energy demand increases by 28%. The more extensive schemes are less influenced by the additional costs, e.g. FrodoKEM requires 9% more energy for the handshake.

5.3 Battery Lifetime Analysis

A Lithium cell (AA, thionyl chloride, 3.6V, 2.6Ah) was used as energy source and a sleep-current of 2.5 μ A was measured to analyze the theoretical battery run time for different KEMs. To compensate aging effects, 70% of the stored energy have been assumed as usable. Furthermore, the energy demand of a complete TLS secured MQTT transaction was considered, including 12 bytes of sensor data which are transmitted to the server. Energy for sensor data acquisition is not included in the estimations, as it strongly depends on the deployed sensor.

Based on different numbers of connections per day, figure 12 shows the theoretic run time estimation for KEMs of all problem families. The energy demand of the SLB-KEMs has been averaged and they are displayed as STRUCT_LVL_X where X denotes the security level for clarification purposes. The chart shows that edge devices with SLB-KEMs can theoretically reach battery run times of 10 years with more than 500 connections per day in hybrid and non-hybrid modes. Edge devices with the high security KEM FrodoKEM, which has almost 20KB of ciphertext and public key, as well as SIKE, which has extensive calculations, can achieve 10 years of battery run time when transferring data more than 100 times per day.

Even though higher security levels and hybrid key exchange have relevant overheads compared to lower security levels and non-hybrid modes, the estimation shows that this overhead is still very low with respect to battery life time.

6 CONCLUSION

In this work, we evaluated the application of post-quantum KEMs to TLS-secured low-power IoT. NIST round 3 finalists and alternate candidates of all three problem families (codes, lattices and isogeny) as well as rank-metric code-based KEMs were implemented into an

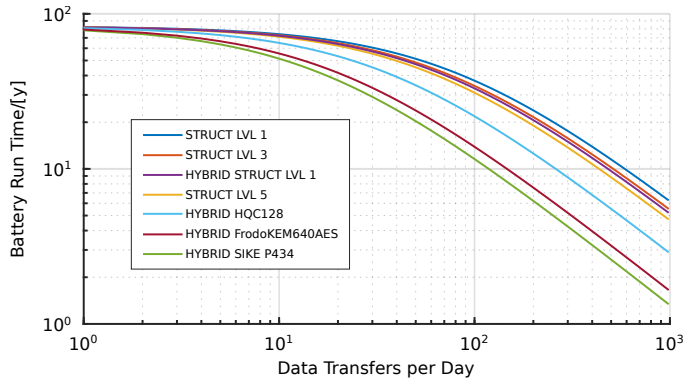


Figure 12: Battery run time estimations for different KEMs

extended mbedTLS library. The costs of quantum computer resistant KEMs, hybrid key exchanges and KEMs with higher security levels were explored and evaluated. Based on our results, we provided estimations for the life time of battery powered IoT edge devices in dependency of the used post-quantum KEM. Our key observations are:

- Contrary to common belief, dedicated cryptographic hardware accelerators are not necessary to facilitate the implementation of energy efficient secure communication in low-power IoT. Against the background of crypto-agile systems, edge devices with KEMs of all problem families can reach more than ten years of theoretical battery run time for typical IoT scenarios.
- The latency-related overhead of code-based and lattice-based post-quantum KEMs is predominantly determined by their larger bandwidth requirements rather than by computational effort. Thus, on top of the previous observation, the possible benefits of dedicated CHAs in terms of energy and latency are very limited in low-power IoT applications, especially for the popular SLB-KEMs. In contrast, SIKE-based handshakes can be significantly improved with CHAs, as their latency and energy demands are mostly driven by calculations even with the highly optimized software implementation from [36].
- Higher NIST security levels cause an overhead of 25% in energy and latency for level 3 KEMs and of 45% in energy and latency for level 5 KEMs.
- Hybrid key exchange can be deployed with a decent overhead to fulfill high governmental security requirements, e.g. for SLB-KEMs it causes an overhead of 13% in latency and of 30% in energy.
- These results go beyond previous reports, proving that the usage of post-quantum KEMs is already feasible today in terms of energy demand and latency for low-power IoT with state-of-the-art hardware, even with hybrid key exchanges, to protect TLS secured communication against QC-based attacks.

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